# Lab 3 : BP Data Cleaning using R

Dataset: Heart Disease dataset

```
heart <- read.csv("heart.csv")
str(heart)
```

```
'data.frame':   1025 obs. of  14 variables:
 $ age     : int  52 53 70 61 62 58 58 55 46 54 ...
 $ sex     : int  1 1 1 1 0 0 1 1 1 1 ...
 $ cp      : int  0 0 0 0 0 0 0 0 0 0 ...
 $ trestbps: int  125 140 145 148 138 100 114 160 120 122 ...
 $ chol    : int  212 203 174 203 294 248 318 289 249 286 ...
 $ fbs     : int  0 1 0 0 1 0 0 0 0 0 ...
 $ restecg : int  1 0 1 1 1 0 2 0 0 0 ...
 $ thalach : int  168 155 125 161 106 122 140 145 144 116 ...
 $ exang   : int  0 1 1 0 0 0 0 1 0 1 ...
 $ oldpeak : num  1 3.1 2.6 0 1.9 1 4.4 0.8 0.8 3.2 ...
 $ slope   : int  2 0 0 2 1 1 0 1 2 1 ...
 $ ca      : int  2 0 0 1 3 0 3 1 0 2 ...
 $ thal    : int  3 3 3 3 2 2 1 3 3 2 ...
 $ target  : int  0 0 0 0 0 0 1 0 0 0 0 ...
```

```
set.seed(123)
n <- nrow(heart)

neg_index <- sample(1:n, 5)
na_index <- sample(setdiff(1:n, neg_index), 5)
high_index <- sample(setdiff(1:n, c(neg_index, na_index)), 5)

heart$trestbps[neg_index] <- -heart$trestbps[neg_index]
heart$trestbps[na_index] <- NA
heart$trestbps[high_index] <- sample(305:350, 5, replace = TRUE)

write.csv(heart, "heart_dirty_data.csv", row.names = FALSE)
```

```
clean_bp <- function(x){
  if(is.na(x)){
    return(NA)
  }
  if(x < 0){
    return(NA)
  }
  if(x > 250){
    return(250)
  }
  return(x)
}

heart$trestbps_clean <- sapply(heart$trestbps, clean_bp)
heart[c(neg_index, na_index, high_index), c("trestbps", "trestbps_clean")]
```

A data.frame: 15 × 2

| | trestbps | trestbps_clean |
|---|---|---|
| | <int> | <dbl> |
| **415** | -170 | NA |
| **463** | -118 | NA |
| **179** | -110 | NA |
| **526** | -130 | NA |
| **195** | -160 | NA |
| **943** | NA | NA |
| **823** | NA | NA |
| **118** | NA | NA |
| **301** | NA | NA |
| **231** | NA | NA |
| **248** | 331 | 250 |
| **14** | 309 | 250 |
| **379** | 331 | 250 |
| **673** | 332 | 250 |
| **610** | 313 | 250 |

```
safe_mean <- function(x){
  result <- tryCatch({
    mean(x, na.rm = TRUE)
  }, error = function(e){
    print("error in mean calculation")
    NA
  })
  return(result)
}

safe_mean(heart$trestbps)
```

```
131.2
```

```
safe_ratio <- function(a, b){
  result <- tryCatch({
    if(is.na(a) | is.na(b)){
      stop("missing value")
    }
    if(b == 0){
      stop("cannot divide by zero")
    }
    a / b
  }, error = function(e){
    print(paste("error:", e$message))
    NA
  })
  return(result)
}

safe_ratio(200, 0)
safe_ratio(200, NA)
safe_ratio(NA, 100)
safe_ratio(200, 100)
```

```
[1] "error: cannot divide by zero"
<NA>
[1] "error: missing value"
<NA>
[1] "error: missing value"
<NA>
2
```

```
heart$ratio <- heart$chol / heart$trestbps_clean
head(heart$ratio, 10)
```

```
1.696 · 1.45 · 1.2 · 1.37162162162162 · 2.1304347826087 · 2.48 · 2.78947368421053 · 1.80625 · 2.075 · 2.34426229508197
```

```
bp <- heart$trestbps

loop_clean <- function(x){
  output <- c()
  for(i in 1:length(x)){
    val <- x[i]
    if(is.na(val)){
      output[i] <- NA
    } else if(val < 0){
      output[i] <- NA
    } else if(val > 250){
      output[i] <- 250
    } else {
      output[i] <- val
    }
  }
  return(output)
}

vector_clean <- function(x){
  x[!is.na(x) & x < 0] <- NA
  x[!is.na(x) & x > 250] <- 250
  return(x)
}

result1 <- loop_clean(bp)
result2 <- vector_clean(bp)

identical(result1, result2)
```

```
TRUE
```

```
t1 <- system.time(for(i in 1:500){ loop_clean(bp) })
t2 <- system.time(for(i in 1:500){ vector_clean(bp) })

t1
t2
```

```
   user  system elapsed
  0.309   0.013   0.323
   user  system elapsed
   0.02    0.00    0.02
```

```
heart$trestbps_clean <- result2
```

```
sum(is.na(heart$trestbps_clean))
min(heart$trestbps_clean, na.rm = TRUE)
max(heart$trestbps_clean, na.rm = TRUE)
mean(heart$trestbps_clean, na.rm = TRUE)
median(heart$trestbps_clean, na.rm = TRUE)

any(heart$trestbps_clean < 0, na.rm = TRUE)
any(heart$trestbps_clean > 250, na.rm = TRUE)
```

```
10
94
250
132.16354679803
130
FALSE
FALSE
```

```
heart$trestbps <- heart$trestbps_clean
heart$trestbps_clean <- NULL

write.csv(heart, "cleaned_heart_data.csv", row.names = FALSE)
```

Conclusion: The vectorized method ran faster than the for loop, since R does not have to repeat the same instruction one element at a time. After cleaning, trestbps has no negative values and nothing above 250, and the 10 missing readings are still marked as NA.